

Hybrid silicon and lithium niobate electro-optical ring modulator

LI CHEN, QIANG XU, MICHAEL G. WOOD, AND RONALD M. REANO*

Electroscience Laboratory, Department of Electrical and Computer Engineering—The Ohio State University, Columbus, Ohio 43212, USA

*Corresponding author: reano.1@osu.edu

Received 24 April 2014; revised 19 June 2014; accepted 25 June 2014 (Doc. ID 210673); published 15 August 2014

Of all oxides, lithium niobate (LiNbO_3) is the gold standard electro-optical material in fiber-optic transmission systems. Modulators based on diffused waveguides in bulk LiNbO_3 substrates are, however, relatively large. In contrast, ring modulators based on silicon-on-insulator are of interest for chip-scale electro-optical modulation, but unstrained crystalline silicon does not exhibit a linear electro-optic effect, so modulation is based on alternative mechanisms such as the plasma dispersion effect. Here, we present a hybrid silicon and LiNbO_3 electro-optical ring modulator operating at gigahertz frequencies. The modulator consists of a 15 μm radius silicon microring and an ion-sliced LiNbO_3 thin film bonded together via benzocyclobutene. Fabricated devices operating in the TE optical mode exhibit an optical loaded quality factor of 14,000 and a resonance tuning of 3.3 pm/V. The small-signal electrical-to-optical 3 dB bandwidth is measured to be 5 GHz. Digital modulation with an extinction ratio greater than 3 dB is demonstrated up to 9 Gb/s. High-speed and low-tuning-power chip-scale modulators that exploit the high-index contrast of silicon with the second-order susceptibility of LiNbO_3 are envisioned. © 2014 Optical Society of America

OCIS codes: (130.3120) Integrated optics devices; (130.3730) Lithium niobate; (130.4110) Modulators; (230.2090) Electro-optical devices; (230.5750) Resonators; (250.7360) Waveguide modulators.

<http://dx.doi.org/10.1364/OPTICA.1.000112>

1. INTRODUCTION

Modulators are fundamental components for optical links. Lithium niobate (LiNbO_3) guided-wave electro-optic modulators satisfy bandwidth, linearity, and chirp requirements in fiber-optic transmission systems [1]. The physical dimensions of diffused waveguide modulators in bulk LiNbO_3 substrates are, however, on the scale of centimeters. The potential for dense integration is limited.

Silicon-on-insulator (SOI) modulators are of interest for chip-scale modulation [2]. Applications include short-reach optical interconnects [3], long-haul optical communications [4], analog optical links [5], and reconfigurable optical filters [6]. While the large refractive index and low optical absorption of silicon make it an attractive medium in the telecommunications wavelength range, unstrained crystalline silicon does not exhibit a linear electro-optic effect. Consequently, silicon optical modulators rely on alternative mechanisms such as the plasma dispersion effect to achieve electro-optical

modulation [7,8]. While effective, plasma dispersion relies on carrier transport. Consequently, refractive index modulation is accompanied by absorption, and steady-state electrical tuning of optical resonances can consume significant power. Alternatively, electro-optic effects from second-order susceptibility can be induced by straining silicon waveguides [9]. Reported values of $V_\pi L$ in the 100 V · cm range indicate, however, that the effect is relatively weak [10,11].

Recently, a hybrid silicon and LiNbO_3 material system consisting of silicon waveguide ring resonators bonded to ion-sliced LiNbO_3 thin films has been introduced to combine the dense integration of silicon photonics with the second-order susceptibility of LiNbO_3 [12–14]. In the hybrid system, a thin film of LiNbO_3 is bonded to the top of a silicon waveguide to serve as a portion of the top cladding of an optical waveguide mode. The advantages are threefold. First, the silicon waveguide can serve as an optically transparent electrode to enhance voltage-induced electric fields in the LiNbO_3 [14].

Second, large shifts of optical resonance with applied voltage are enabled in the absence of significant absorption. Third, steady-state DC power consumption for the tuning of optical resonance frequencies can potentially be very low due to the capacitive geometry. Tunable filters and radio-frequency electric field sensors have been demonstrated; however, a high-speed electro-optical ring modulator has not been reported in the literature to date.

In this paper, we present the experimental demonstration of a hybrid silicon and LiNbO₃ electro-optical microring resonator modulator operating at gigahertz frequencies. The device consists of a 15 μm radius silicon ring resonator and a 1 μm thick z-cut LiNbO₃ ion-sliced thin film bonded together by benzocyclobutene (BCB). Fabricated devices operating in the transverse-electric (TE) optical mode exhibit an optical loaded quality factor of 14,000 and a resonance tuning of 3.3 pm/V. High-frequency scattering parameters are used to extract an RC circuit model for the modulator. The small-signal electrical-to-optical 3 dB bandwidth is measured to be 5 GHz. Digital modulation with an extinction ratio (ER) greater than 3 dB is demonstrated up to 9 Gb/s.

This paper is organized as follows. Design and fabrication details of the modulator are first described. Next, DC and small-signal high-frequency measurements are presented. High-speed data modulation results are then conveyed, followed by concluding remarks.

2. METHODS

A. Design

A schematic of the hybrid silicon and LiNbO₃ modulator is shown in Fig. 1. The cross section is through the center of the ring resonator. The device consists of a 15 μm radius silicon rib waveguide ring and a 1 μm thick z-cut ion-sliced LiNbO₃ thin film bonded via BCB [13]. The rib waveguides are 500 nm wide, with a 45 nm slab thickness and 205 nm rib height. The silicon slab is patterned so that it exists only around and exterior to the ring. The silicon core and surrounding silicon slab layer are doped to function as a transparent conductor with reduced series resistance [14]. A voltage applied between the top electrode and the bottom electrode produces an electric field confined between the top electrode and the silicon waveguide core. The electric field interacts with the portion of the optical mode in the LiNbO₃ cladding, modifying the mode effective index by the linear electro-optic effect.

As a result, the optical transmission response of the ring resonator is modulated.

The 1550 nm wavelength TE optical mode distribution is shown in Fig. 2 calculated by the beam propagation method. The refractive indices of silicon, SiO₂, LiNbO₃, and BCB are set to 3.48, 1.44, 2.21, and 1.54, respectively, in the simulation. The effective index of the optical mode is 2.66. Also shown is the voltage-induced electric field (yellow vectors) from a DC voltage applied between the top electrode and the silicon transparent conductor, using material permittivities in finite element method calculations [15–17].

The optical mode and the electric field vectors overlap in the LiNbO₃. The fraction of the optical mode power in the LiNbO₃ is 11%. The TE mode accesses the r_{13} electro-optic coefficient in LiNbO₃ ($r_{13} = 8 \text{ pm V}^{-1}$ in bulk LiNbO₃) [15] and takes advantage of the nearly vertical voltage-induced electric field in the LiNbO₃.

The device speed is limited by the RC time constant and the photon lifetime in the ring resonator [2]. The electrical resistance originates primarily from the silicon transparent conductor and the contact resistance. To reduce the resistance, the silicon waveguide is blanket implanted with P-type dopants at a light dose, followed by a P-type heavy dose on the slab. The heavily doped region is 300 nm away from the silicon core to avoid excessive optical loss.

B. Fabrication

The fabrication process begins with a SOI wafer with a silicon device layer thickness of 250 nm and a buried SiO₂ layer thickness of 1 μm . The silicon waveguides are patterned with hydrogen silsesquioxane (HSQ) resist using electron beam lithography and inductively coupled plasma reactive ion etching (ICP-RIE) to obtain a rib waveguide geometry with a slab thickness of 45 nm [18]. The coupling gap between the bus waveguides and the ring waveguides is 200 nm. The slab is patterned with a second HSQ resist layer and ICP-RIE. After patterning, the slab only exists around and exterior to the ring waveguides and on one side of the bus waveguides. After patterning the slab regions, the HSQ mask over the waveguide and the slab is removed, and 20 nm of plasma-enhanced chemical vapor deposition (PECVD) SiO₂ is deposited. The sample is then blanket implanted with 45 keV BF₂⁺ ions at a fluence of $1.8 \times 10^{13} \text{ ions cm}^{-2}$ to lightly dope the silicon

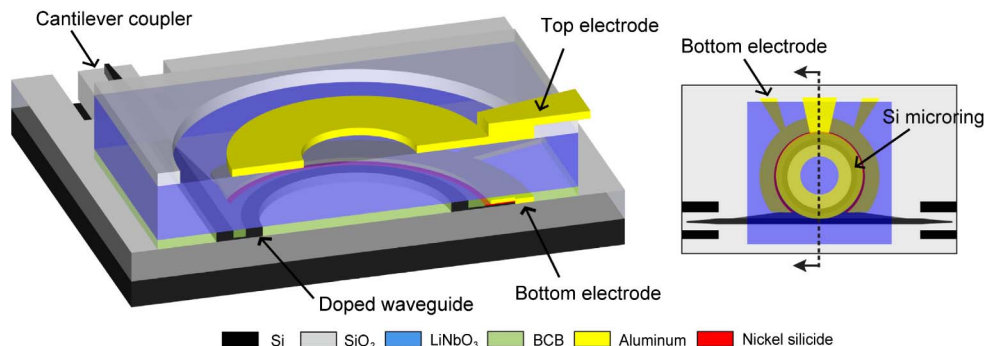


Fig. 1. Schematic of the hybrid silicon and LiNbO₃ ring modulator.

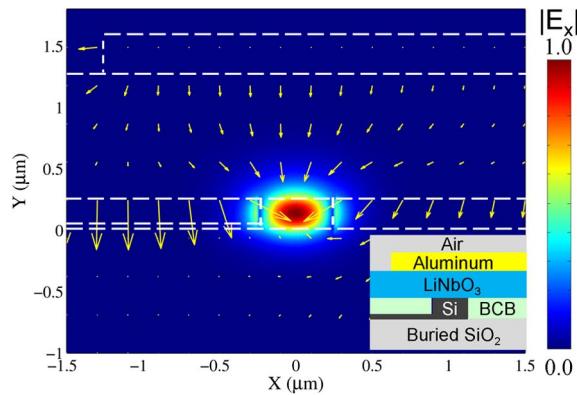


Fig. 2. Calculated optical TE mode distribution at 1550 nm wavelength (E_x component) and DC voltage-induced electric field vectors.

core. After implantation, the sample is annealed using rapid thermal annealing (RTA) to activate and drive in the dopants.

The slab is then heavily doped with 45 keV BF_2^+ ions at a fluence of $3 \times 10^{15} \text{ cm}^{-2}$. Figure 3(a) shows the top-view scanning electron micrograph of the device after this step. Nickel silicide is formed in the heavily doped region before 100 nm of aluminum is deposited as the bottom electrode. A 350 nm thick BCB layer is then spin-coated and etched back to a 250 nm thickness to allow for planarization of the waveguide topology.

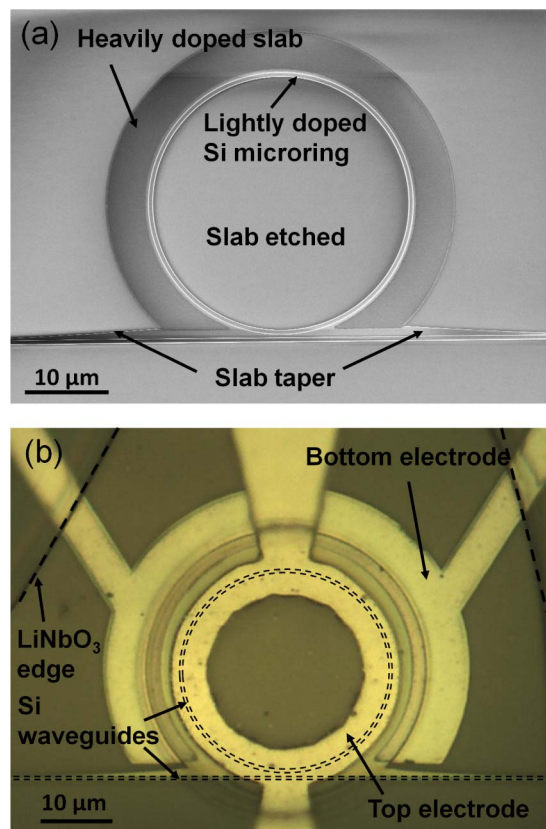


Fig. 3. (a) Scanning electron micrograph of the silicon microring resonator after slab patterning and doping; (b) top-view optical micrograph of the fabricated device.

To obtain LiNbO_3 thin films, a z-cut LiNbO_3 wafer is implanted with He^+ ions with an implantation energy of 342 keV and a fluence of $4 \times 10^{16} \text{ ions cm}^{-2}$. After implantation, the wafer is flip bonded to an unpolished silicon wafer and baked on a hotplate at 300°C for 5 min. The heating causes blistering of the implanted LiNbO_3 wafer surface due to the aggregation of helium bubbles [19]. One micrometer thick LiNbO_3 thin films are exfoliated from the bulk and transferred to the silicon wafer. The exfoliated thin films on the silicon wafer are then annealed by RTA at 1000°C for 30 s to repair the crystal lattice and restore the electro-optic properties. The surface roughness of the exfoliated side of the LiNbO_3 thin film is 6 nm. LiNbO_3 thin films are transferred and bonded to the Si microring resonators using a micro-vacuum tip on a probe station. After transfer, the sample is annealed to cure the BCB. Residual BCB not covered by the LiNbO_3 thin film is etched via ICP-RIE with CF_4/O_2 chemistry. One micrometer thick PECVD SiO_2 film is deposited as a capping layer. The SiO_2 film over the ring resonator is removed to form a 40 μm diameter via hole. A 300 nm thick top aluminum electrode, connected to a ground-signal-ground radio-frequency (RF) pad, is patterned on top of the LiNbO_3 thin film. Finally, cantilever couplers are patterned for fiber-to-chip optical coupling [20,21]. A top-view optical micrograph of the fabricated device is shown in Fig. 3(b).

3. RESULTS AND DISCUSSION

A. DC and Small-Signal High-Frequency Measurements

Optical transmission measurements are performed to characterize the electrical tuning of optical resonances. TE-mode light from a tunable infrared continuous-wave laser source is coupled through the input and output fiber-to-chip cantilever couplers of the modulator. DC voltage is applied to the top electrode of the modulator while the bottom electrode is grounded. Figure 4 shows the measured TE-mode spectrum as a function of the applied DC voltage. The resonance wavelength redshifts with increasingly positive voltage. The total optical insertion loss is 4.3 dB. The insertion loss is from

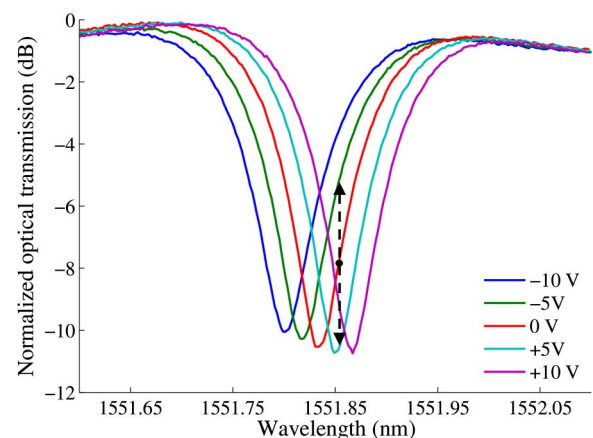


Fig. 4. Measured optical transmission of a single resonance as a function of applied voltage.

3 dB of fiber-to-waveguide coupling loss and 1.3 dB of waveguide transmission loss [18,20,21]. The decrease of the optical transmission minimum with voltage is attributed to Fabry–Perot fringes that are present due to fiber-to-chip coupling. The resonance shift is 66 pm for a change in DC bias from –10 to 10 V, indicating 3.3 pm/V tuning, which is equivalent to a $V_\pi L$ of 9.1 V · cm. The measured quality factor is 14,000, the FWHM is 13.7 GHz, and the group index is 4.0. At 1551.856 nm, the optical transmission intensity varies by 5.2 dB with a –5 to 5 V voltage swing, as indicated by the dashed arrow in Fig. 4.

The RF scattering parameter, S_{11} , is measured with a 20 GHz vector network analyzer operating in a 50 Ω system. Figure 5(a) shows the measured S_{11} magnitude and phase data. The magnitude of S_{11} drops sharply to –2 dB from 40 MHz to 2 GHz and then gradually rolls off to –4.5 dB at 20 GHz. An RC circuit model extracted from the measured S_{11} is shown in Fig. 5(b) [8]. Parameter R_C denotes the resistance of the metal electrodes. Parameter C_P represents the parasitic capacitance between the metal electrodes through the top dielectric materials and the air. Parameters C_J and R_S model the electrical path through the silicon transparent electrode and the LiNbO₃ layer. Parameters C_{OX} and R_{Si} model the electrical path through the buried oxide and the silicon substrate. The value of C_J is in good agreement with finite element method simulations. The relatively high value of R_S indicates a lower than expected doping concentration in the waveguide and a higher than expected contact resistance [22]. The relatively large C_P value is mainly attributed to the parasitic capacitance of the electric pads. The sharp drop at the lower frequency band is attributed to the RF coupling through the substrate.

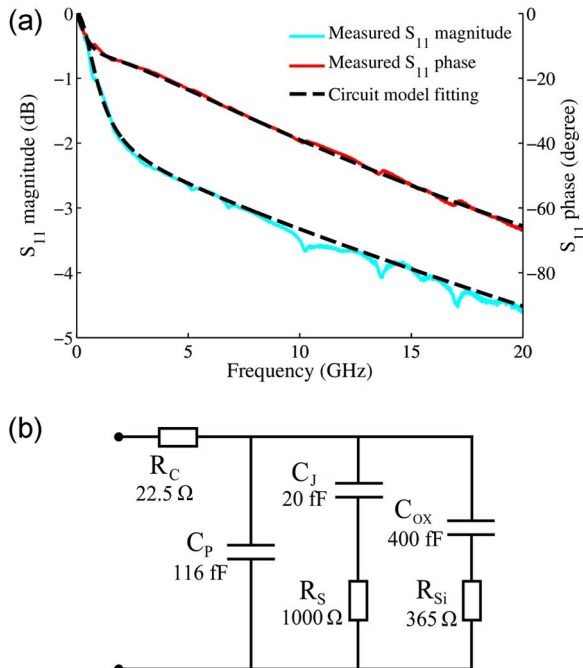


Fig. 5. (a) RF S_{11} scattering parameter; (b) RC circuit model of the modulator.

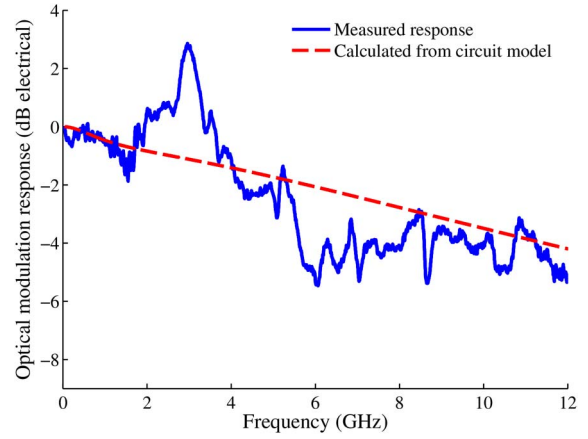


Fig. 6. Electrical-to-optical modulation response.

The small-signal electrical-to-optical modulation response is shown in Fig. 6. The modulated optical signal is detected with a photodetector with a 3 dB bandwidth of 12 GHz. Also shown in Fig. 6 is the frequency response of the voltage across C_J in the circuit model shown in Fig. 5(b) [8]. The measured 3 dB optical response bandwidth of the modulator is approximately 5 GHz. While the general trend of the optical responses matches the circuit model, the optical response contains additional dips and peaks across the entire frequency band.

The dips and peaks on the optical modulation response are a result of acousto-optic resonances [23–25]. The electric fields between the electrodes excite acoustic waves in the 1 μ m thick LiNbO₃, producing resonances in the gigahertz frequency range. Multiple resonance peaks apparent in the S_{11} measurements at 1.7 GHz and corresponding harmonics indicate the conversion of the electrical energy to acoustic energy at the acoustic resonances. Consequently, refractive index changes occur in the LiNbO₃ through the elasto-optic effect. The acoustic resonances can be suppressed by roughening the surface of the piezoelectric material to increase the scattering loss of the acoustic wave [26,27]. The surface roughness can be produced by dry or wet etching [28]. Sufficiently thick LiNbO₃ avoids optical mode interaction with the rough surface.

B. High-Speed Digital Modulation

A schematic for the high-speed digital modulation characterization is shown in Fig. 7 [8]. A pulse pattern generator (PPG) outputs a $2^{31} - 1$ pseudorandom-bit-stream data stream (PRBS31) with a voltage swing from –1 to 1 V. The peak-to-peak amplitude of the signal is amplified to 5 V (–2.5 to 2.5 V) with a modulator driver amplifier. Due to the microwave reflection from the capacitive device, the voltage swing across the modulator electrode is doubled to 10 V at DC [8]. The output light from the modulator is passed through a fiber amplifier and an optical passband filter with 0.5 nm bandwidth to amplify the signal and suppress the amplified spontaneous emission noise. The amplified optical signal is attenuated and connected to a 30 GHz optical module on

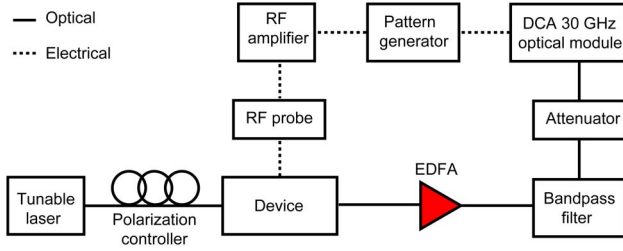


Fig. 7. Measurement setup for digital characterization. EDFA, erbium-doped fiber amplifier.

a digital communication analyzer (DCA) synchronized to the clock of the PPG for generating optical eye diagrams. The optical bias wavelength is tuned to maximize the ER.

The left column of Fig. 8 shows measured optical eye diagrams at 1, 4.5, 5, and 9 Gb/s, with ERs of 4.7, 4.5, 4, and 3 dB, respectively. The right column of Fig. 8 shows simulated eye diagrams at the corresponding bit rates with

ideal PRBS driving signals. The modeling incorporates the measured frequency response in Fig. 6 and the optical resonance spectrum of the modulator in Fig. 4. The ER and shape of the simulated optical eyes agree well with the measurement results. In particular, the amplitude ringing for the 1 Gb/s eye and the partial eye closing on the left part of the 5 Gb/s eye closely match the simulation.

The total energy consumption of the modulator is estimated to be 4.4 pJ/bit at 9 Gb/s with a 10 V swing. The energy per bit encompasses 0.4 pJ/bit from C_J , 1.8 pJ/bit from C_P , and 2.2 pJ/bit from C_{OX} . The energy consumption on C_J is only 9% of the total energy consumption. The majority of the energy consumption is attributed to charging and discharging of the large area electrical pads for test and measurement. Energy consumption can be reduced by optimizing device capacitances. Operating in the optical TM mode also reduces the energy per bit due to the increase in tunability from a larger electro-optic coefficient ($r_{33} = 31 \text{ pm V}^{-1}$ [15]) and greater overlap between RF and optical fields.

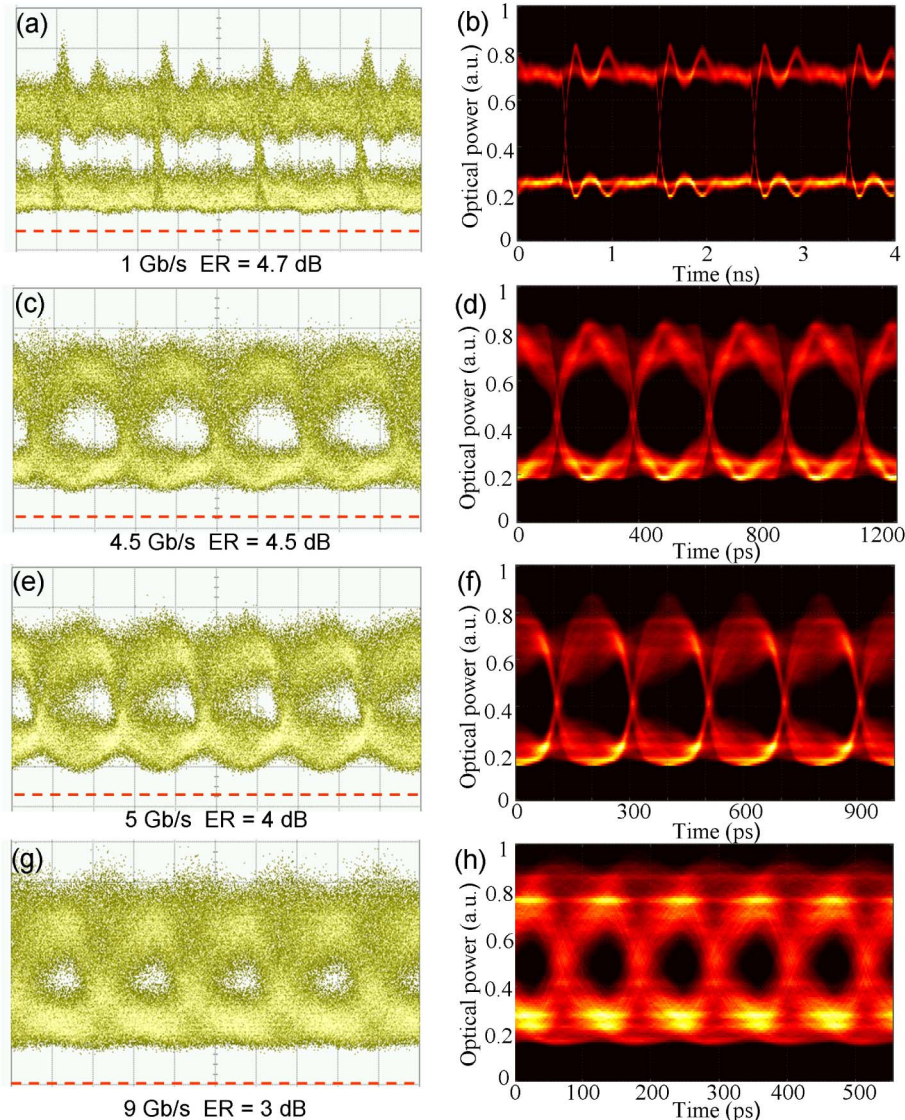


Fig. 8. Measured (left column) and simulated (right column) optical eye: (a), (b) 1 Gb/s; (c), (d) 4.5 Gb/s; (e), (f) 5 Gb/s; (g), (h) 9 Gb/s. The red dashed line in the measurement indicates the reference level for zero optical input. The vertical scale is 500 μW per division.

Steady-state DC power consumption for the tuning of optical resonance frequencies is ideally zero in the capacitive device.

4. CONCLUSION

A hybrid silicon and LiNbO₃ microring resonator modulator operating at gigahertz frequencies is designed, fabricated, and characterized. The hybrid modulator consists of an ion-sliced LiNbO₃ thin film bonded to a silicon waveguide ring resonator via BCB. Fabricated devices operating in the TE optical mode exhibit an optical loaded quality factor of 14,000 and a resonance tuning of 3.3 pm/V. High-frequency scattering parameters are used to extract an RC circuit model for the modulator. The small-signal electrical-to-optical 3 dB bandwidth is measured to be 5 GHz. Digital modulation with an ER greater than 3 dB is demonstrated up to 9 Gb/s. Future work involves optimizing the RC time constant, suppressing the acousto-optic resonances, and reducing the driving voltage in TE- and TM-mode designs to achieve higher data rates and lower energy consumption. High-speed and low-tuning-power chip-scale optical modulators that exploit the high-index contrast of silicon with the second-order susceptibility of LiNbO₃ are envisioned.

More broadly, new horizons become apparent when exploiting the capability of silicon to provide submicrometer spatial confinement of light and the ability of lithium niobate to mediate second-order nonlinear optical effects. Empowering silicon with second-order susceptibility opens a suite of nonlinear optics to the chip scale, including second-harmonic generation, difference frequency generation, optical rectification, and sum frequency generation for applications in classical and quantum information processing [29–31].

FUNDING INFORMATION

Army Research Office (ARO) (W911NF-12-1-0488).

REFERENCES

1. E. L. Wooten, K. M. Kissa, A. Yi-Yan, E. J. Murphy, D. A. Lafaw, P. F. Hallemeier, D. Maack, D. V. Attanasio, D. J. Fritz, G. J. McBrien, and D. E. Bossi, "A review of lithium niobate modulators for fiber-optic communications systems," *IEEE J. Sel. Top. Quantum Electron.* **6**, 69–82 (2000).
2. G. T. Reed, G. Mashanovich, F. Y. Gardes, and D. J. Thomson, "Silicon optical modulators," *Nat. Photonics* **4**, 518–526 (2010).
3. D. A. B. Miller, "Device requirements for optical interconnects to silicon chips," *Proc. IEEE* **97**, 1166–1185 (2009).
4. A. Biberman, S. Manipatruni, N. Ophir, L. Chen, M. Lipson, and K. Bergman, "First demonstration of long-haul transmission using silicon microring modulators," *Opt. Express* **18**, 15544–15552 (2010).
5. M. Streshinsky, A. Ayazi, Z. Xuan, A. E.-J. Lim, G.-Q. Lo, T. Baehr-Jones, and M. Hochberg, "Highly linear silicon traveling wave Mach-Zehnder carrier depletion modulator based on differential drive," *Opt. Express* **21**, 3818–3825 (2013).
6. S. S. Djordjevic, L. W. Luo, S. Ibrahim, N. K. Fontaine, C. B. Poitras, B. Guan, L. Zhou, K. Okamoto, Z. Ding, M. Lipson, and S. J. B. Yoo, "Fully reconfigurable silicon photonic lattice filters with four cascaded unit cells," *IEEE Photon. Technol. Lett.* **23**, 42–44 (2011).
7. Q. Xu, B. Schmidt, S. Pradhan, and M. Lipson, "Micrometre-scale silicon electro-optic modulator," *Nature* **435**, 325–327 (2005).
8. G. Li, X. Zheng, J. Yao, H. Thacker, I. Shubin, Y. Luo, K. Raj, J. E. Cunningham, and A. V. Krishnamoorthy, "25 Gb/s 1V-driving CMOS ring modulator with integrated thermal tuning," *Opt. Express* **19**, 20435–20443 (2011).
9. R. S. Jacobsen, K. N. Andersen, P. I. Borel, J. Fage-Pedersen, L. H. Frandsen, O. Hansen, M. Kristensen, A. V. Lavrinenko, G. Moulin, H. Ou, C. Peucheret, B. Zsigri, and A. Bjarklev, "Strained silicon as a new electro-optic material," *Nature* **441**, 199–202 (2006).
10. B. Chmielak, M. Waldow, C. Matheisen, C. Ripperda, J. Bolten, T. Wahlbrink, M. Nagel, F. Merget, and H. Kurz, "Pockels effect based fully integrated, strained silicon electro-optic modulator," *Opt. Express* **19**, 17212–17219 (2011).
11. B. Chmielak, C. Matheisen, C. Ripperda, J. Bolten, T. Wahlbrink, M. Waldow, and H. Kurz, "Investigation of local strain distribution and linear electro-optic effect in strained silicon waveguides," *Opt. Express* **21**, 25324–25332 (2013).
12. Y. S. Lee, G.-D. Kim, W.-J. Kim, S.-S. Lee, W.-G. Lee, and W. H. Steier, "Hybrid Si-LiNbO₃ microring electro-optically tunable resonators for active photonic devices," *Opt. Lett.* **36**, 1119–1121 (2011).
13. L. Chen and R. M. Reano, "Compact electric field sensors based on indirect bonding of lithium niobate to silicon microrings," *Opt. Express* **20**, 4032–4038 (2012).
14. L. Chen, M. G. Wood, and R. M. Reano, "12.5 pm/V hybrid silicon and lithium niobate optical microring resonator with integrated electrodes," *Opt. Express* **21**, 27003–27010 (2013).
15. K. K. Wong, *Properties of Lithium Niobate* (INSPEC, 2002).
16. M. E. Mills, P. Townsend, D. Castillo, S. Martin, and A. Achen, "Benzocyclobutene (DVS-BCB) polymer as an interlayer dielectric (ILD) material," *Microelectron. Eng.* **33**, 327–334 (1997).
17. S. M. Sze and K. K. Ng, *Physics of Semiconductor Devices* (Wiley, 1981).
18. M. G. Wood, L. Chen, J. R. Burr, and R. M. Reano, "Optimization of electron beam patterned hydrogen silsesquioxane mask edge roughness for low-loss silicon waveguides," *J. Nanophoton.* **8**, 083098 (2014).
19. A. Guarino, G. Poberaj, D. Rezzonico, R. Degl'Innocenti, and P. Günter, "Electro-optically tunable microring resonators in lithium niobate," *Nat. Photonics* **1**, 407–410 (2007).
20. M. Wood, P. Sun, and R. M. Reano, "Compact cantilever couplers for low-loss fiber coupling to silicon photonic integrated circuits," *Opt. Express* **20**, 164–172 (2012).
21. P. Sun and R. M. Reano, "Cantilever couplers for intra-chip coupling to silicon photonic integrated circuits," *Opt. Express* **17**, 4565–4574 (2009).
22. R. Ding, T. Baehr-Jones, Y. Liu, R. Bojko, J. Witzens, S. Huang, J. Luo, S. Benight, P. Sullivan, J.-M. Fedeli, M. Fournier, L. Dalton, A. Jen, and M. Hochberg, "Demonstration of a low V_{π} modulator with GHz bandwidth based on electro-optic polymer-clad silicon slot waveguides," *Opt. Express* **18**, 15618–15623 (2010).
23. J. L. Nightingale, R. A. Becker, R. C. Willis, and J. S. Vrhel, "Characterization of frequency dispersion in Ti-diffused lithium niobate optical devices," *Appl. Phys. Lett.* **51**, 716–718 (1987).
24. R. L. Jungerman and C. A. Flory, "Low-frequency acoustic anomalies in lithium niobate Mach-Zehnder interferometers," *Appl. Phys. Lett.* **53**, 1477–1479 (1988).
25. W. K. Burns, M. M. Howerton, R. P. Moeller, R. Krähenbühl, R. W. McElhanon, and A. S. Greenblatt, "Low drive voltage, broad-band LiNbO₃ modulators with and without etched ridges," *J. Lightwave Technol.* **17**, 2551–2555 (1999).
26. A. Vorobiev, J. Berge, S. Gevorgian, M. Löffler, and E. Olsson, "Effect of interface roughness on acoustic loss in tunable thin film bulk acoustic wave resonators," *J. Appl. Phys.* **110**, 024116 (2011).
27. M. Yoshida, S. Okamoto, T. Omiya, K. Kasai, and M. Nakazawa, "256 QAM digital coherent optical transmission using Raman amplifiers," *IEICE Trans. Commun. E* **94-B**, 417–424 (2011).

28. Z. Ren, P. J. Heard, J. M. Marshall, P. A. Thomas, and S. Yu, "Etching characteristics of LiNbO₃ in reactive ion etching and inductively coupled plasma," *J. Appl. Phys.* **103**, 034109 (2008).
29. M. Cazzanelli, F. Bianco, E. Borga, G. Pucker, M. Ghulinyan, E. Degoli, E. Luppi, V. Vénier, S. Ossicini, D. Modotto, S. Wabnitz, R. Pierobon, and L. Pavesi, "Second-harmonic generation in silicon waveguides strained by silicon nitride," *Nat. Mater.* **11**, 148–154 (2011).
30. M. Wächter, C. Matheisen, M. Waldow, T. Wahlbrink, J. Bolten, M. Nagel, and H. Kurz, "Optical generation of terahertz and second-harmonic light in plasma-activated silicon nanophotonic structures," *Appl. Phys. Lett.* **97**, 161107 (2010).
31. K. Banaszek, A. B. U'Ren, and I. A. Walmsley, "Generation of correlated photons in controlled spatial modes by down conversion in nonlinear waveguides," *Opt. Lett.* **26**, 1367–1369 (2001).